

Artificial Intelligence

For HEDSPI Project

Lecturer 10 – First Order Logic

Lecturers :

Dr. **Le Thanh Huong**

Dr. Tran Duc Khanh

Dr. Hai V. Pham

HUST

First Order Logic

- Syntax
- Semantic
- Inference
 - Resolution

First Order Logic (FOL)

- First Order Logic is about
 - Objects
 - Relations
 - Facts
- The world is made of objects
 - *Objects* are things with individual identities and properties to distinguish them
 - Various *relations* hold among objects. Some of these relations are functional
 - Every fact involving objects and their relations are either *true* or *false*

FOL

- Syntax
- Semantic
- Inference
 - Resolution

FOL Syntax

■ Symbols

- Variables: $x, y, z, \dots$
- Constants: $a, b, c, \dots$
- Function symbols (with arities): $f, g, h, \dots$
- Relation symbols (with arities): p, r, r
- Logical connectives: $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$
- Quantifiers: $\exists, \forall$

FOL Syntax

- Variables, constants and function symbols are used to build terms
 - $X, \text{Bill}, \text{FatherOf}(X), \dots$
- Relations and terms are used to build predicates
 - $\text{Tall}(\text{FatherOf}(\text{Bill})), \text{Odd}(X), \text{Married}(\text{Tom}, \text{Marry}), \text{Loves}(Y, \text{MotherOf}(Y)), \dots$
- Predicates and logical connective are used to build sentences
 - $\text{Even}(4), \forall X. \text{Even}(X) \Rightarrow \text{Odd}(X+1), \exists X. X > 0$

FOL Syntax

■ Terms

- Variables are terms
- Constants are terms
- If $t_1, \dots, t_n$ are terms and f is a function symbol with arity n then $f(t_1, \dots, t_n)$ is a term

FOL Syntax

■ Predicates

- If $t_1, \dots, t_n$ are terms and p is a relation symbol with arity n then $p(t_1, \dots, t_n)$ is a predicate

FOL Syntax

■ Sentences

- True, False are sentences
- Predicates are sentences
- If α, β are sentences then the followings are sentences

$$\exists x.\alpha, \forall x.\alpha, (\alpha), \neg\alpha, \alpha \wedge \beta, \alpha \vee \beta, \alpha \Rightarrow \beta, \alpha \Leftrightarrow \beta$$

FOL Formal grammar

Sentence	::=	AtomicS ComplexS
AtomicS	::=	True False RelationSymb(Term,...) Term = Term
ComplexS	::=	(Sentence) Sentence Connective Sentence $\neg$ Sentence Quantifier Sentence
Term	::=	FunctionSymb(Term,...) ConstantSymb Variable
Connective	::=	$\wedge$ $\vee$ $\rightarrow$ $\leftrightarrow$
Quantifier	::=	$\forall$ Variable $\exists$ Variable
Variable	::=	a b ... x y ...
ConstantSymb	::=	A B ... <i>John</i> 0 1 ... π ...
FunctionSymb	::=	F G ... <i>Cosine</i> <i>Height</i> <i>FatherOf</i> $+$...
RelationSymb	::=	P Q ... <i>Red</i> <i>Brother</i> <i>Apple</i> $>$...

FOL

- Syntax
- Semantic
- Inference
 - Resolution

FOL Semantic

- Variables
 - Objects
- Constants
 - Entities
- Function symbol
 - Function from objects to objects
- Relation symbol
 - Relation between objects
- Quantifiers
 - $\exists x.P$ true if P is true under some value of x
 - $\forall x.P$ true if P is true under every value of x
- Logical connectives
 - Similar to Propositional Logic

FOL Semantic

- Interpretation (D, σ)
 - D is a set of objects, called *domain* or *universe*
 - σ is a mapping from variables to D
 - C^D is a member of D for each constant C
 - F^D is a mapping from D^n to D for each function symbol F with arity n
 - R^D is a relation over D^n for each relation symbol R with arity n

FOL Semantic

- Given an interpretation (D, σ) , semantic of a term/sentence α is denoted

$$[\alpha]_a^D$$

- Interpretation of terms

$$[x]_\sigma^D := \sigma(x)$$

$$[C]_\sigma^D := C^D$$

$$[F(t_1, \dots, t_n)]_\sigma^D := F^D([t_1]_\sigma^D, \dots, [t_n]_\sigma^D)$$

FOL Semantic

■ Interpretation of sentence

$$\begin{aligned}
 \llbracket R(t_1, \dots, t_n) \rrbracket_{\sigma}^D &:= \text{True} && \text{iff } \langle \llbracket t_1 \rrbracket_{\sigma}^D, \dots, \llbracket t_n \rrbracket_{\sigma}^D \rangle \in R^D \\
 \llbracket \neg \varphi \rrbracket_{\sigma}^D &:= \text{True/False} && \text{iff } \llbracket \varphi \rrbracket_{\sigma}^D = \text{False/True} \\
 \llbracket \varphi_1 \vee \varphi_2 \rrbracket_{\sigma}^D &:= \text{True} && \text{iff } \llbracket \varphi_1 \rrbracket_{\sigma}^D = \text{True} \text{ or } \llbracket \varphi_2 \rrbracket_{\sigma}^D = \text{True} \\
 \llbracket \exists x \varphi \rrbracket_{\sigma}^D &:= \text{True} && \text{iff } \llbracket \varphi \rrbracket_{\sigma'}^D = \text{True for some } \sigma' \text{ the} \\
 \llbracket \varphi_1 \wedge \varphi_2 \rrbracket_{\sigma}^D &:= \llbracket \neg(\neg \varphi_1 \vee \neg \varphi_2) \rrbracket_{\sigma}^D \\
 \llbracket \varphi_1 \rightarrow \varphi_2 \rrbracket_{\sigma}^D &:= \llbracket \neg \varphi_1 \vee \varphi_2 \rrbracket_{\sigma}^D \\
 \llbracket \varphi_1 \leftrightarrow \varphi_2 \rrbracket_{\sigma}^D &:= \llbracket (\varphi_1 \rightarrow \varphi_2) \wedge (\varphi_2 \rightarrow \varphi_1) \rrbracket_{\sigma}^D \\
 \llbracket \forall x \varphi \rrbracket_{\sigma}^D &:= \llbracket \neg \exists x \neg \varphi \rrbracket_{\sigma}^D
 \end{aligned}$$

Example

■ Symbols

- Variables: $x, y, z, \dots$
- Constants: $0, 1, 2, \dots$
- Function symbols: $+, *$
- Relation symbols: $>, =$

■ Semantic

- Universe: $\mathbb{N}$ (natural numbers)
- The meaning of symbols
 - Constants: the meaning of 0 is *the number zero*, ...
 - Function symbols: the meaning of $+$ is *the natural number addition*, ...
 - Relation symbols: the meaning of $>$ is *the relation greater than*, ...

FOL Semantic

- Satisfiability
 - A sentence α is satisfiable if it is true under some interpretation (D, σ)
- Model
 - An interpretation (D, σ) is a model of a sentence α if α is true under (D, σ)
 - Then we write $(D, \sigma) \models \alpha$
- A sentence is valid if every interpretation is its model
- A sentence α is valid in D if $(D, \sigma) \models \alpha$ for all σ
- A sentence is unsatisfiable if it has no model

Example

- Consider the universe N of natural numbers
 - $\exists x. x+1 > 5$ is satisfiable
 - $\forall x. x+1 > 0$ is valid in N
 - $\exists x. 2x+1 = 6$ is unsatisfiable